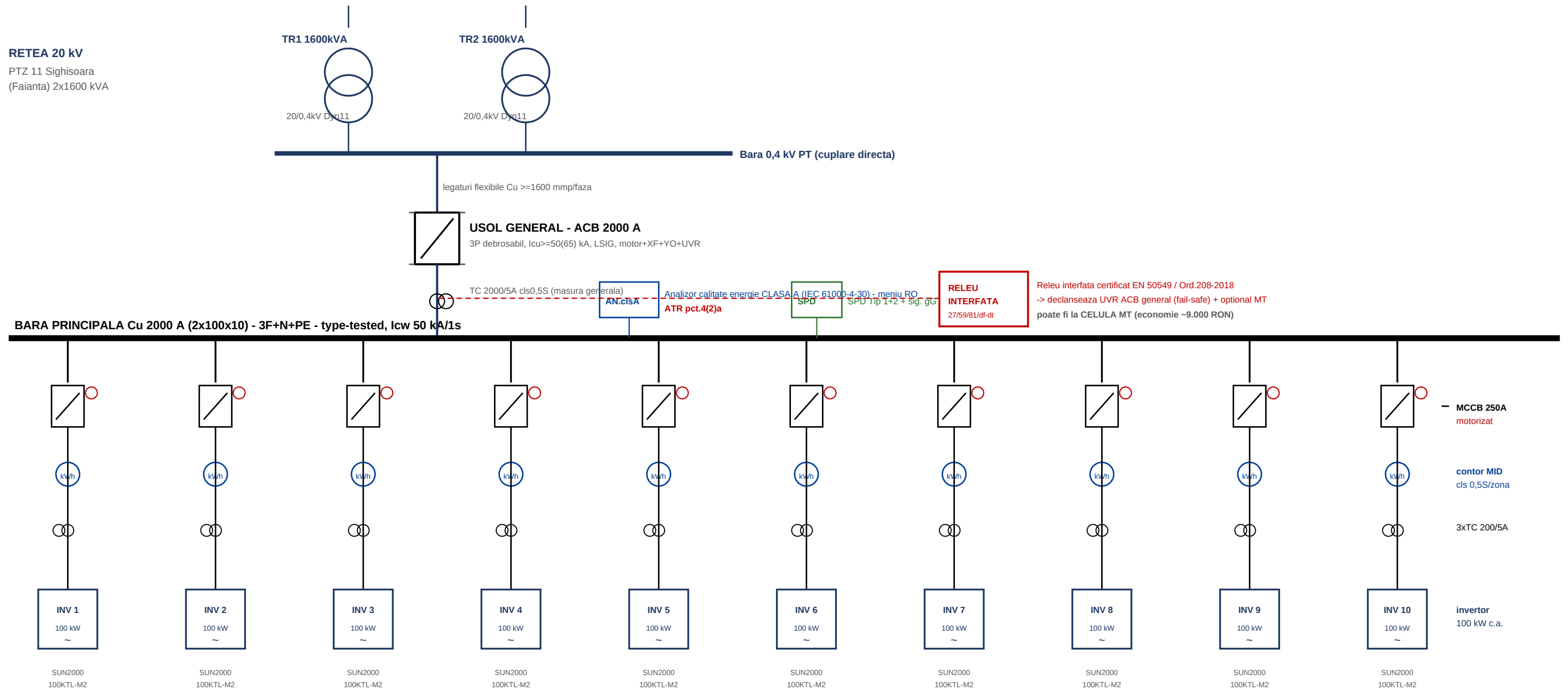


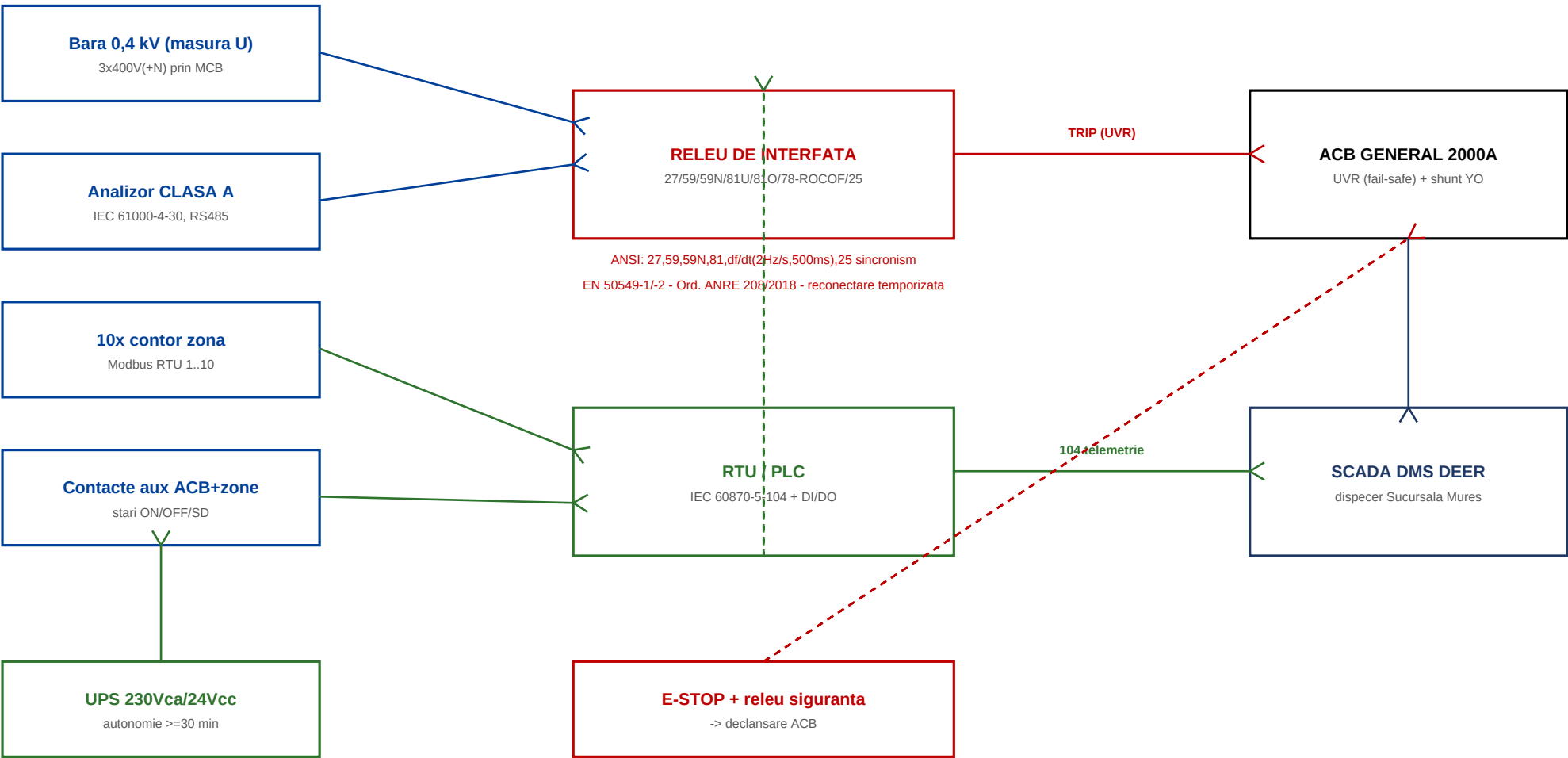
SCHEMA MONOFILARA (single-line) - TEG-PV 1 MVA



STOCARE: LUNA2000-200KWH-2H1 (200 kW) - racordata pe bara PV / inverter dedicat,
conform Ord. ANRE 3/2023 (scope furnizor PV Suntime)

SCADA/RTU IEC 60870-5-104 -> dispecer DEER Mures (P,Q,U,I,f + pozitie intrerupator); UCMT/sir cleme SCADA;
magistrala Modbus RS485 contoare zone -> gateway -> switch fibra. UPS 24Vcc mentine releu interfata + comanda.

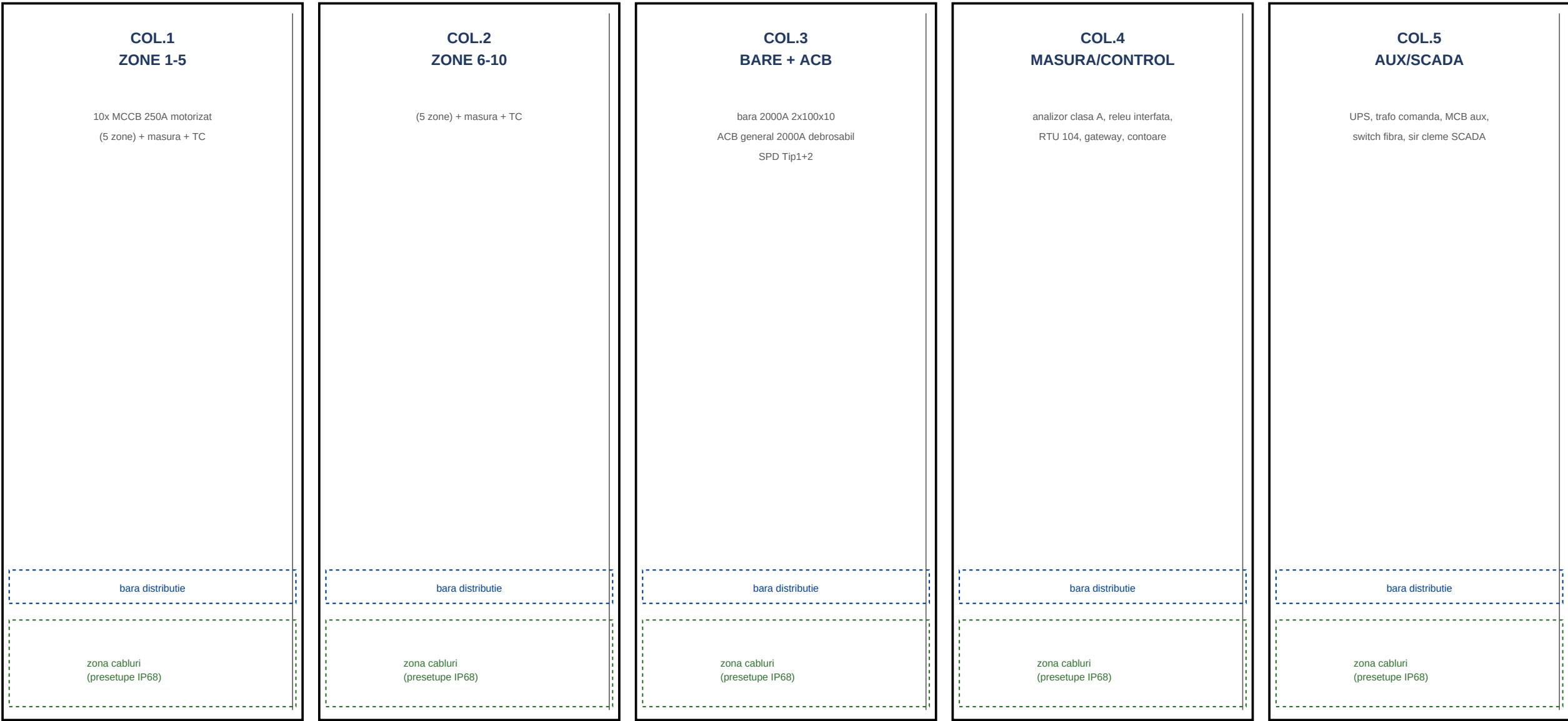
SCHEMA FUNCTIONALA - PROTECTIE DE INTERFATA + COMANDA / TELEMETRIE



Principiu: pierderea retelei / abatere U-f-df/dt -> releul interfata comanda UVR-ul ACB general (declansare fail-safe, sigura si la bara moarta).
Invertoarele Huawei au protectie proprie EN 50549; releul de interfata este decuplarea unica a centralei. RTU trimite telemetria la dispecer DEER (IEC 60870-5-104).

NOTA: daca CELULA MT de delimitare are deja releu numeric certificat (67/67N/df-dt), releul de interfata JT poate fi omis - decizie in PTE-IU / aviz CTE-Z DEER.

DISPUNERE COLOANE + FORMA DE SEPARARE (Form 3b) - orientativ pt. asamblare



Carcasa: dulap podea IP54, ~2200x2000x600 mm, Form 3b (bare separate de aparate, borne separate) - sistem TYPE-TESTED EN 61439.

Termic: ventilatie fortata cu filtru + termostat, anticondens. Ancorare seismica soclu. Iluminat + priza service RCD.

Legare la pamant: bara PE continua, braiduri echipotentiale usi, priza de pamant coordonata cu grila PT (Rp<=1 ohm).

Intrari: feederi 10x N2XH 3x95+50 prin presetupe IP68 in COL.1/2; iesire flexibila 2000A/faza spre bara PT din COL.3.

AVERTIZARI OBLIGATORII PE USA: 'ATENTIE - DUBLA ALIMENTARE (retea + PV)', pericol arc electric (arc-flash), schema mimica,

placuta de identificare IEC 61439-2 (In, Icw, Ipk, IP, tensiuni, producatorul ansamblului, an), identificare faze.

Verificari de rutina la asamblare: continuitate PE, incercare dielectrica, verificare cablaj+functionala (raport EN 61439-2).